

Project Status Report

Project Name: Detecting Disease Contacts

Team Name: Circuit Breakers

Project Manager: Robin Pottathuparambil

Team Members: Tarod Anderson, Jared Beyers, Ronil Doshi, Isaiah Maberry, Jennifer Tapia

Report Date: September 11, 2025

Reporting Period: August 18, 2025 to September 11, 2025

Management Summary

Defined milestones completed: 6 of 6 (100%)
Defined tasks completed: 0 of 0 (0%) of child tasks in work breakdown structure
Total estimated project hours used: 200 of 1600 hours (12.5%)
Ahead of (or Behind) schedule by: 0 labor hours, 0 days
Known defects: N/A
Staff members on project: N/A
Contingency hours remaining: 100% of 350 hours

Schedule

Initial estimated completion date: April 9, 2026
Previous estimated completion date: N/A
Current estimated completion date: April 9, 2026

Key Milestones Table

ID	Title	Planned Completion Date	Previous Forecast Completion Date	Current Forecast Completion Date	Actual Completion Date
1	Formed Team	8/21/25	8/21/25	8/21/25	8/21/25
2	Selected Project	8/28/25	N/A	8/28/25	8/28/25
3	Drafted Functional Requirements	9/12/2025	N/A	9/12/2025	
4	Drafted Non-Functional Requirements	9/12/2025	N/A	9/12/2025	
5	Status Report (9/11/2025)	9/11/2025	N/A	9/11/2025	9/11/2025
6	Established an initial budget	9/11/2025	N/A	9/11/2025	9/11/2025

Product Size

Potential Microcontroller suggestion: ESP32-CAM

SPI Flash: Default 32Mbit

RAM: 520KB SRAM +4M PSRAM

Effort

Life Cycle Activity	This Reporting Period (labor-hours)		Project to Date (labor-hours)	
	Planned Effort	Actual Effort	Planned Effort	Actual Effort
Requirements Draft	20	20	20	20
Status Report (9/11/2025)	20	20	20	20

Cost

Life Cycle Activity	This Reporting Period		Project to Date	
	Planned Cost	Actual Cost	Planned Cost	Actual Cost
Materials	900	0	900	0
Cloud computing	100	0	100	0

Requirements Status

I D	Functional Requirements	Team Member Responsible	Effort (%)	Verification	Completed (%)
F R 1	The system should detect all persons within declared range	Jared Beyers	60%	Testing	0 %
F R 2	Obtain real-time calculation for distance using an ultrasonic sensor	Jared Beyers	15%	Analysis	0 %
F R 3	Detect objects crossing a set distance	Jared Beyers	15%	Analysis	0 %
F R 4	The system should track one person across the field of motion (recognizing it is the same person)	Ronil Doshi	30%	Inspection	0 %

F R 5	The system should communicate with a mobile app and display information	Jennifer Tapia	45 %	Demonstration	0 %
F R 6	Use cloud computation to apply algorithm on sensor data to create listable contact instance	Ronil Doshi	20%	Demonstration	0 %
F R 7	The system should record how long any given person stays within the threshold distance	Tarod Anderson	32%	Demonstration	0 %
F R 8	The system should detect and store data based frequency of contacts	Isaiah Maberry	30%	Analysis	0 %
F R 9	The system should have the capability to track multiple people within a range of 2 m.	Ronil Doshi	40%	Demonstration	0 %
F R 10	The system should increment a data array with timestamp every time someone leaves the threshold range	Tarod Anderson	31%	Analysis	0 %
F R 11	The system should pair the duration, location, and closest approach of contact into a contact instance	Tarod Anderson	32%	Analysis	0 %
F R 12	The mobile app should display a searchable list of contact instances	Jennifer Tapia	30 %	Demonstration	0 %
F R 13	Cloud System should receive 5 updates from the camera and sensors a second	Isaiah Maberry	50%	Testing	0 %

ID	Non-Functional Requirements	Team Member Responsible	Effort (%)	Verification	Completed (%)
NFR 1	The system should be wearable	Tarod Anderson	5%	Inspection	0 %
NFR 2	The system should be discrete	Jared Beyers	10	Inspection	0 %
NFR 3	The system should use a battery for 12-hour operation	Isaiah Maberry	20%	Analysis	0 %
NFR 4	Mobile application should be user friendly	Jennifer Tapia	25 %	Demonstration	0 %
NFR 5	The system should be under 4.5KG	Ronil Doshi	10%	Inspection	0 %

Top Five Risks

N/A

Open Issues

N/A

Action Items

- Finalize requirements
- Start specifications

Defects

N/A